

Contribution journal (Canvas item 7)

Canvas S7: "Upload detailed proposal (reflection journal) of what you accomplished; each team member talks about contribution in the contribution journal" **Canvas R4:** "Individual contribution: no entry minus 20, no participation minus 100"

1. My role

- This project was conducted individually. I carried out each method decision, reviewed and signed each checkpoint, wrote my interpretation, and directed the AI agents that implemented, ran and checked the codebase, collaborated with them, and asked the tutor-reviewer agent clarifying questions.

2. What I did, with proof

Deliverable	What I did	Proof (file and commit hash)
1 dataset URL document	Recorded the Kaggle source and URL for the stroke dataset	docs/dataset_url.md, 89193d0
2 one page proposal	wrote the one page proposal from my signed decisions.	docs/MT_JosephClay_ITAI1371_Proposal.pdf, 2d6c658
3 split notebook	Chose and signed the 70/30 split (D1: plain random, seed 33), then reviewed the split checks	notebooks/01_load_split.ipynb, D1 card 149de94, notebook 5f0abd5
4 EDA on train only	Wrote the observation under every chart and the findings cell, then fixed 3 points the data validator raised	notebooks/02_eda_train.ipynb, a24f16f, fixes c17a968
5 before and after notebooks	signed D2 to D8, ran notebooks 03 and 04, wrote all H3 notes, and asked for the stroke by work_type cell.	notebooks/03_preprocess_before_after.ipynb, notebooks/04_smote_balance.ipynb, a7ddbcb, 1f4ae1e, c3594f5
6 .ipynb files	Reviewed each notebook at its checkpoint	notebooks/, H1 78e6825, H2 ea395d1
7 journals	Wrote both journals	docs/MTJournal_C_JosephClay_ITAI1371.pdf, docs/MTJournal_R_JosephClay_ITAI1371.pdf, 2d6c658
8 final clean dataset	reviewed the final files and signed the integrity checks at H3.	data/processed/stroke_clean_final.csv, a7ddbcb

3. Decisions I made

Card	My choice (short)	ADR
D1 split	A: plain random 70/30, seed 33, no stratify	docs/adr/0001-d1-split-plain-random-70-30.md

Card	My choice (short)	ADR
D2 rows and columns	Remove the 1 gender "Other" row from train; drop id before modeling	docs/adr/0002-d2-remove-other-row-drop-id.md, 550d8a9
D3 missing values	Fill bmi with the train median (28.1); add a bmi_missing flag first; keep smoking "Unknown" as a category	docs/adr/0003-d3-median-fill-with-missing-flag.md, 550d8a9
D4 encoding	Full one hot with handle_unknown ignore; work_type encoded after SMOTE	docs/adr/0004-d4-full-one-hot-encoding.md, 550d8a9
D5 scaling and normalization	StandardScaler on age, glucose and bmi; MinMax shown as the normalization exercise	docs/adr/0005-d5-standard-scaler-on-number-columns.md, 550d8a9
D6 order around SMOTE	Fill, encode, scale, then plain SMOTE	docs/adr/0006-d6-fill-encode-scale-then-smote.md, 550d8a9
D7 SMOTE setup	Plain SMOTE on work_type, k 5, "auto", random_state 33, columns cast to float first	docs/adr/0007-d7-plain-smote-settings.md, 550d8a9
D8 after SMOTE and final file	Round 0/1 and argmax repair, both numeric checks, scaled units, is_synthetic flag	docs/adr/0008-d8-repair-check-and-final-file.md, 550d8a9

4. AI use disclosure

Tools used: Kiro CLI with a builder agent and the five project subagents (notebook-runner, data-validator, work-verifier, repo-auditor, tutor-reviewer). I also used an AI assistant (Grok Bot) to explain concepts and options.

What AI did:

- The Kiro builder agent wrote the Python code (notebook cells and functions) from the decisions I signed on the decision cards, and drafted the ADRs from those cards for me to review and save.
- Between my checkpoints, the notebook-runner, data-validator, work-verifier and repo-auditor ran the notebooks, checked them and each other, and wrote a one page summary for each checkpoint. At H2 they flagged two items for me: the builder read one file outside the project folder (a kernel settings file), and notebook 01 prints test shares. I kept the test shares because the D1 card's checks require them.
- At H3 the builder made the float change, and the work-verifier then raised it as escalation E3 for my review because it had chosen a SMOTE input setting (cast to float first) that my D7 card left open. I reviewed both options and signed float first, so blended 0/1 values get rounded at 0.5 instead of being cut off.
- The tutor-reviewer explained concepts, like how a random seed works, and asked me questions.
- The AI assistant explained each data chart, recommended an option for D2 to D8 with pros and cons, and drafted the wording of my EDA observations and decision card text from the numbers in my notebooks.

What I did: I chose and signed every decision card, approved or corrected each checkpoint summary, reviewed and edited every observation and card before saving it, set the decision values in config.py, saved the ADRs, wrote both journals and the proposal, and made my own commits, pushes and checkpoint tags.

How I checked AI output:

- The data-validator rechecked 84 claims in my EDA notes. It found 0 wrong but 3 that needed fixing: a vague bullet about id, a reference to "box 6" that didn't match any label, and a notebook title hidden by an HTML comment. I fixed all three (c17a968).
- When I filled in config.py, the ruff check stopped my commit because two lines were too long. I shortened them, and the decision slot tests passed 15 of 15 (ccc2dc4).

Learning and preparation note:

- Before building, I used a reference website as an all inclusive guide to understand the new concepts, mapped against the assignment criteria and made to fit my learning style. With that understanding, I built my own repo, made and signed every decision, and ran every notebook individually, so every number, decision and conclusion here comes from my own run. This project was also my chance to apply multi agent orchestration, which I learned from that reference, and I count it as part of my learning process and individual contribution.

5. Time log (optional)

Date	Checkpoint	Hours	What
Before Oct 4	Prep	Several sessions over multiple days	Learning the concepts with the midterm explainer, mapped to the assignment criteria
Before Oct 4	H1	Several sessions	Setting up the repo, the agents and protocols, and signing H1
Sun Oct 4	H2 to H4	All day, into the evening	H2 to H4: EDA notes, signing D1 to D8, notebooks 03 and 04, ADRs, journals, submit